EPM-Bench: Prospective Memory for Embodied Agents, Adjudicated from World State

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Abstract

Tell a household agent "when you notice the kettle boiling, turn it off" — and then cancel the request. Will the agent still act? Agent memory research is almost entirely retrospective: no existing benchmark gives an embodied agent a deferred instruction and adjudicates the full intention lifecycle — acquisition, retention, execution, and *termination* — from world state, and no prior work isolates the execution of cancelled intentions. We introduce EPM-Bench, a dual-simulator benchmark (AI2-THOR kitchens; Habitat apartments) of 7,564 procedurally generated, frozen episodes in which a seeded event engine fires triggers outside the agent's field of view, every episode ships with matched control / live / cancelled twins, and deterministic checkers grade outcomes from simulator state alone — no LLM judge anywhere. Across mechanical reference agents ($n \geq 48$/cell) and a real-LLM pilot, commission errors emerge as an *architectural* property: a naively injected memory store executes 46–78% of cancelled intentions while a typed store with mechanical revocation bookkeeping eliminates violations entirely, at comparable execution rates; the ranking is invariant across model scale and both simulators. A window-bounded mock actor decomposes failures into information starvation versus attention tax, and a grading-regime ablation shows that selection-style probes pass 52.9% of agents that already violated in-world — world-state adjudication is the only regime that grades the record rather than the disposition. We release environments, generator, frozen suite, checkers, and validity probes.

1 Introduction

Embodied agents are beginning to hold obligations over time. A household robot told "if you ever see the toaster on, switch it off" must carry that intention through minutes of unrelated work, notice a trigger that fires behind its back, act on it — and, just as importantly, *stop* carrying it once the user says "never mind." Human cognition treats this faculty as its own memory system: *prospective memory* (PM), the memory for delayed intentions, with a psychology literature spanning monitoring costs, cue focality, intention load, and *commission errors* — executing an intention after it has been cancelled.

Agent memory research has flourished, but almost entirely in the *retrospective* direction: what did the agent see, where are objects, what happened yesterday. Benchmarks for embodied memory (FindingDory, MEMENTO, WorldLines) ask agents to recall the past; text-side PM benchmarks (PM-Bench, TriggerBench) grade deferred instructions by question answering or menu selection rather than by what the agent actually does to the world. No prior work measures whether an embodied agent executes an intention its user has revoked. The gap has a structural cause: QA-style and selection-style grading express an agent's *disposition* at probe time; a commission error lives in the *history* of the world. Only world-state adjudication can see it (§7).

We make four contributions:

1. **EPM-Bench**, the first execution-graded prospective-memory benchmark in 3D worlds: deferred instructions injected mid-task, a seeded event engine that changes the world out of view, per-step visibility instrumentation, matched cancellation twins, and deterministic world-state checkers (§3).

2. **EAC**, executed-after-cancellation, as a first-class metric, validated by deaf-agent probes and commonsense controls (§3.4).

3. **Findings**: commission is an architectural property of the memory pathway, not a property of model scale, intention load, or simulator (§4); scheduling, not storing, is the hard problem, with a three-way cost/execution/compliance trade-off that exists already at the information layer (§6).

4. **A mock-actor methodology** that separates information starvation from attention tax, letting mechanical floors be established before any LLM token is spent (§6).

2 Related Work

Prospective memory in text agents. PM-Bench introduces deferred intentions over long dialogues; TriggerBench tests trigger-action associations. Both grade by text answers or option selection, and neither places the trigger in a world the agent must perceive. ProEvent maintains intention records at the data level. None measures post-cancellation execution.

Timing and proactivity. Gaia2, SentinelBench, and SimuHome test whether agents act at the right time in simulated schedules; they grade timeliness of live intentions, not the termination half of the lifecycle.

Embodied memory. FindingDory, MEMENTO, and WorldLines benchmark retrospective memory in simulators; integrated memory systems (RoboMemory, eMEM) ship spatial/episodic/semantic/temporal stores. Across 26 systems and benchmarks we surveyed, no memory architecture contains a prospective store, and no benchmark contains a cancellation condition.

Termination neighbours. VIGIL touches the completion half of obligation management; Revisable-style in-flight replanning handles goal changes mid-execution; Chameleon stores observation–action delays at the control level. The strict question — does a revoked intention stay dead? — remains unmeasured.

Psychology anchor. Our design imports the multiprocess framework (monitoring vs. spontaneous retrieval), cue focality, and commission-error paradigms (including post-cancellation persistence under prior implementation intentions) from three decades of human PM research.

3 EPM-Bench

3.1 Design principles

(1) Full lifecycle in one episode. Acquisition (a deferred instruction arrives in the briefing, mid-task), retention (interference from ongoing chores), execution (the trigger fires out of view; noticing must be earned by looking), termination (matched cancelled variants; exactly-once completion). **(2) World-state adjudication only.** Success and violation are computed from simulator event logs by deterministic code; no LLM judge. **(3) Memory isolated from common sense.** Every seed has a no-instruction control twin; we report memory contribution $\mathrm{MEM} = \mathrm{EXEC}_{\mathrm{live}} - \mathrm{EXEC}_{\mathrm{control}}$. **(4) Noticing separated from acting.** Visibility is instrumented per step (field of view, distance, occlusion), so every failure is classified encoding / noticing / execution / termination.

3.2 Environments

Kitchen tier (AI2-THOR). Object-state triggers: a toaster or faucet is toggled on behind the agent's back; execution means the device is actually off in the final world state. Foreground chores are pick-and-place errands. Ten kitchen scenes audited solvable (oracle 100%). **Apartment tier (habitat-sim).** Multi-room navigation; the trigger is a marker appearing out of view; execution requires arriving *and declaring* (no accidental pass-by completions). Visibility is geometric: FOV, distance, and line-of-sight.

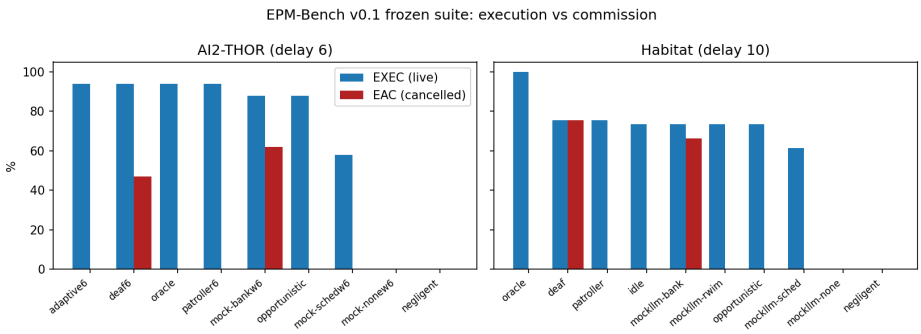


Figure 1: Headline cell of the frozen suite (THOR dev, n=50/arm). No store: 0% execution. Naive always-injected store: 88% execution but 78%/46% executed-after-cancellation (early/late arms). Typed store with mechanical strikethrough and 6-step resurfacing: zero violations on both arms, at 58% execution (the resurfacing beat-gap). Controls confirm MEM: 88/58 points of true memory contribution.

3.3 Episodes, twins, and difficulty axes

An episode is a tuple (scene, seed, chores, intentions, event schedule, cancellation schedule, variant). Four matched variants per seed — control / live / cancel-early / cancel-late — differ only in briefing and cancellation timing. Difficulty axes, all validated in pilot: event delay (3–14 steps; execution falls 91.7→25.0% for scripted agents), intention load (1–3 concurrent intentions), cue focality (trigger on vs. off the chore route: 88.5 vs. 40%), cancellation timing, scene scale (kitchen vs. apartment), and perception tier (privileged text vs. RGB).

Frozen suite v0.1. 7,564 episodes (6,000 THOR across a dev kitchen and three held-out kitchens; 1,564 Habitat across an apartment and a held-out castle), 1,891 complete twin groups, SHA-256 manifest. Sampled oracle audits pass on every scene; a same-seed double run produces bit-identical event logs.

3.4 Metrics

EXEC: fraction of live intentions executed (state change / arrive-and-declare), exactly once. **EAC** (headline): executed-after-cancellation, strictly after the revocation was delivered. A deaf probe agent that cannot receive cancellations calibrates the metric (its strict EAC equals its EXEC on early-cancel arms, 94/94% in THOR); honest agents score 0. **MEM**: EXEC(live) − EXEC(control), the memory contribution net of common sense (a Haiku agent fixes a burning toaster it was never asked about 30% of the time; the control twin removes this). **NOTICE**: first-visible-step vs. execution-step decomposition; failure attribution shares. **COST**: steps, injected notes/tokens, foreground chores completed.

4 Experiments: mechanical layer ($n \geq 48$/cell)

4.1 The scripted frontier

Reference agents (oracle, negligent, opportunistic, periodic patroller, adaptive patroller, deaf) reproduce the signature phenomena of human PM mechanically: monitoring has a foreground price and space sets the price (periodic patrolling is strictly dominated in the apartment tier); *where* to look beats *how often* (coverage-greedy vantage selection lifts execution 70.8→95.8% at equal cost); incidental noticing decays with delay while disciplined monitoring degrades slowly; intention load degrades everything (95.8/83.3/54.2% for loads 1/2/3); and a one-line adaptive rule (reset the monitoring clock on incidental sightings) collapses the whole fixed-frequency frontier to a single dominating point.

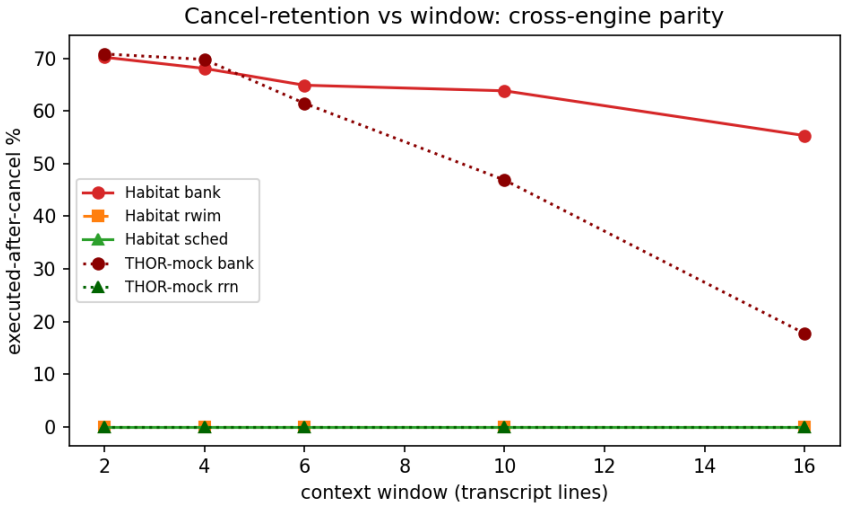


Figure 2: Cancel-retention half-life across engines. Naive store EAC falls with window size; revocation-bookkeeping controls are pinned at zero at every window, in both simulators.

4.2 The memory-policy triptych, on the frozen suite

A window-bounded rule actor — it can act only on text inside its context window — makes the information boundary mechanical (Fig. 1): with no store the briefing evicts and execution is 0% (window size is a dose: 0/0/29/63/67% for w=3..48); an always-injected store pins the intention but lets the *revocation* scroll away (88% EXEC, 78/46% EAC); a typed store that strikes cancelled intentions and announces the strike once achieves 0% EAC at every window size. The cancel-retention half-life is monotone in window size in both engines (THOR: 71→18% EAC for w=2..16; Habitat: 70→55%) while the strikethrough control sits at exactly 0% everywhere (Fig. 2). Commission is window eviction plus schedule persistence — reproducible without any model.

4.3 Generalization

On three held-out THOR kitchens the scripted ladder is unchanged (95.8%), the naive store drops to 68.1% EXEC / 38.9% EAC and the typed store to 48.6% / 0% — rankings invariant, absolutes scene-dependent. In the held-out Habitat castle, patrolling collapses 75.5→29.2% (the spatial-scale law) while oracle remains 100%.

5 Experiments: LLM tier (pilot; solidification grid pending)

Numbers in this section are from the August pilot (n=8–20 per arm, Claude Haiku / Sonnet); the $n\geq$50, multi-family grid is scripted and costed ($128) and blocked only on API access. ***[TODO: replace with grid results]***

Commission is real and architectural. With a naive always-injected store, Haiku executes cancelled intentions 70% of the time in the kitchen and 60% in the apartment; with the typed scheduled store, violations fall to the common-sense floor (10%) and 0% respectively, at one third of the token cost.

Scale does not buy prospective memory. Sonnet's no-store execution (37.5%) exactly equals its no-intention control: apparent memory is all common sense. Commission softens but persists (60→37.5%); the ranking of memory policies is unchanged (Fig. 3).

LLM-delegated scheduling fails in structured ways. A naive LLM-coroutine memory manager emits 28–35 notes per 40 steps, all present-tense chore micromanagement, never the deferred intention (10% execution); a scoped variant relabels chores as "deferred intentions" and leaves the real one unguarded

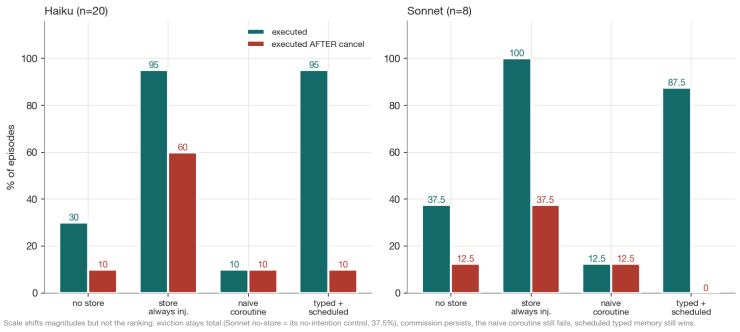


Figure 3: Model scale does not change the memory-policy ranking (pilot).

(30%); actors clock out with obligations open. Mechanical resurfacing restores execution; uniform resurfacing collapses under load (90/30/20% for 1/2/3 intentions) — *which* intention to resurface, when, is the open scheduling problem this benchmark poses.

6 Mechanism: information floor vs. attention tax

Because the mock actor has no attention limits, whatever it loses is information starvation; whatever a real LLM loses beyond that is attention tax. At load 3 the mechanical execution floor under full injection is 0.74 (every lost episode traces to a trigger that never entered the field of view); the real Haiku scheduled agent sits at 0.20–0.30 — a 40–50 point attention tax that no injection schedule can explain. Sight-triggered injection is mechanically optimal (matches full injection at one fifth the cost) yet collapses behaviorally: retrieval at cue onset arrives too late without prior activation — the machine edition of the implementation-intention effect. Sparse injection starves execution but also leaks fewer commissions: cost, execution, and compliance pull in three directions before any model enters the room.

7 Grading-regime ablation: only world state sees the record

We grade the *same* 1,440 cancelled intentions (654 in-world violations) three ways, giving end-of-episode probes exactly the actor's information. A selection-style "would you act on it now?" probe passes **52.9%** of violators — their disposition is clean by episode end; the violation lives in the history. A QA-style "was it cancelled?" quiz passes no violators but false-alarms on **63.4%** of clean episodes (starved schedules leave nothing to recite). World-state adjudication is the only regime that grades what happened rather than what the agent would say (Fig. 4).

8 Discussion and safety

An agent that executes revoked intentions is not merely forgetful; it is disobedient in a way invisible to dialogue-level evaluation — the interface says "okay" while the world says otherwise. EAC gives this failure a number. We view prospective memory as a measurable, safety-relevant capability axis: the same bookkeeping that zeroes EAC also makes the agent's obligations inspectable by humans.

9 Limitations

LLM-tier numbers are pilot-scale pending the costed grid ([**TODO:** $n \geq 50$**, 2–3 model families, prompt ablations, RGB at scale**]); intentions are single-user and explicit (no inferred intentions); navigation is teleport-based in the kitchen tier; both tiers are simulators — our claims live at the memory/decision layer, and sim-to-real transfer of low-level control is out of scope. The late-cancel arm in THOR coincides with the fastest scripted response (v0.2 will enforce a one-step gap).

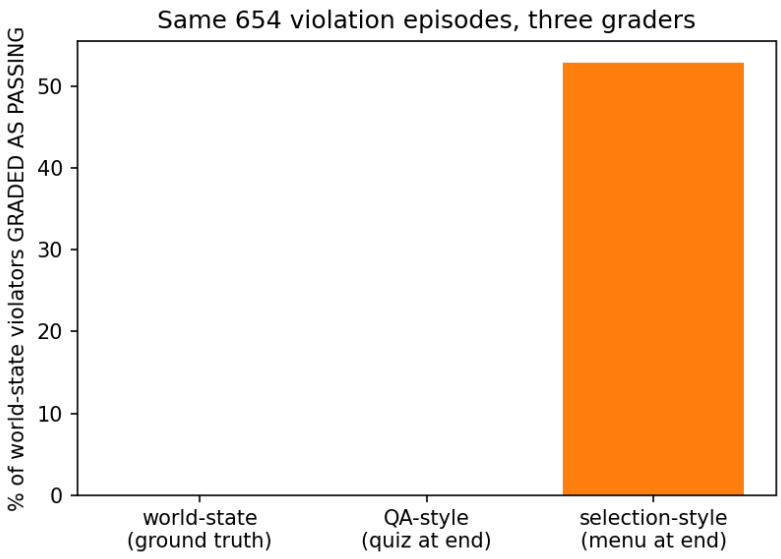


Figure 4: Three graders, same episodes. Selection probes are blind to history; QA probes false-alarm on clean episodes; world-state grading is the only faithful regime.

10 Conclusion

EPM-Bench turns "remember to act — and remember to stop" into a measured, reproducible quantity. The benchmark's central lesson is that storing is easy and scheduling is the science: commission errors are manufactured or eliminated by the memory pathway's bookkeeping, independent of model scale. We release the frozen suite, checkers, and probes, and propose EAC as a standard reporting axis for agent memory.

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